

Efficient Dual-Pathway Valorization of Lignin-Derived VA via Biosensor-Assisted Directed Evolution

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Abstract:

Lignin is an abundant renewable source of aromatic carbon, yet its structural heterogeneity severely limits biological valorization. A three-step biological funneling strategy was established to converge diverse lignin-derived aromatics into vanillic acid (VA), providing a unified starting point for downstream engineering. Building on previous experimental evidence from our laboratory supporting 4-HB (4-HB)-to-chorismate (CHA) conversion, the pathway was extended upstream by investigating VA-to-4-HB conversion, with the low catalytic activity of the demethoxylation enzyme (ODM) identified as a key bottleneck. By integrating error-prone PCR with a PobR-based 4-HB-responsive transcriptional biosensor, variant 1-20-7 was identified. Its measured activity was 29.28-fold that of the wild type(WT) upon rescreening. Meanwhile, a VA-to-Pyruvate (Pyr) pathway was constructed toward fully green bioconversion. Finally, a low-cost, modular microfluidic platform was developed to support future high-throughput screening. Together, molecular evolution, pathway engineering, and hardware-assisted screening provide a framework for the valorization of lignin-derived carbon.

Keywords: lignin valorization; demethoxylation; directed evolution; biosensor-assisted screening; microfluidics

1. Introduction

Lignocellulosic biomass is one of the most abundant renewable resources on Earth, yet only a small fraction is currently converted into high-value products. While numerous bioconversion routes have been established for cellulose and hemicellulose, lignin remains exceptionally recalcitrant because of its heterogeneous, cross-linked, and chemically stable aromatic architecture^[1-3]. This evolutionarily established recalcitrance severely restricts the entry of lignin-derived aromatic carbon into efficient biomanufacturing. Breaking this natural barrier and enabling controllable biological utilization of heterogeneous aromatic carbon therefore remains a central challenge in lignin valorization^[4,5].

Substantial advances have extended lignin valorization from catalytic depolymerization to biological funneling; however, most biological strategies remain constrained by native catabolic logic. Diverse lignin-derived aromatics are typically funnelled into protocatechuate (PCA) or catechol (CA), followed by aromatic ring cleavage and assimilation into central metabolism^[4,5,7]. Even state-of-the-art funneling systems, capable of converting diverse lignin-derived aromatics to PCA at up to 99.89%, remain challenged by substrate toxicity and metabolic flux imbalance^[6]. Thus, the frontier of lignin valorization is no longer simply whether lignin can be utilized, but whether its aromatic carbon can escape native degradation and be redirected toward new high-value biosynthetic destinations.

Building on established chemical strategies for lignin depolymerization^[1-3], this study bridges chemical bond cleavage with de novo biological synthesis. Four structurally distinct lignin-derived aromatic monomers were targeted for conversion into VA through a multienzyme funneling strategy^[8-10]. A non-natural VA→4-HB→CHA route was subsequently proposed. Experimental evidence supporting 4-HB-to-CHA conversion had previously been obtained in our laboratory; therefore, the missing upstream VA-to-4-HB connection was investigated in the present study. Native VA metabolism predominantly channels VA toward PCA through LigM- or VanAB-mediated O-demethylation^[11,12], whereas no characterized enzyme has been reported to directly catalyze VA→4-HB demethoxylation. Rather than reinforcing native degradation, this strategy therefore rewires the metabolic fate of lignin-derived aromatic carbon toward CHA, a central biosynthetic branch point with broad downstream potential^[13].

To further advance toward fully green bioconversion, a VA→Pyr pathway was constructed to reduce dependence on externally supplied Pyr^[7]. In parallel, a self-developed, low-cost microfluidic platform integrating biosensing, fluorescence detection, and droplet sorting was established to enable sensitive, high-throughput exploration of substantially larger mutant libraries^[14,15]. Collectively, aromatic-carbon funneling, non-natural pathway construction, directed evolution, green pathway engineering, and hardware-assisted high-throughput screening were integrated into a unified framework that expands the biological boundaries of lignin valorization(**Figure 1**).

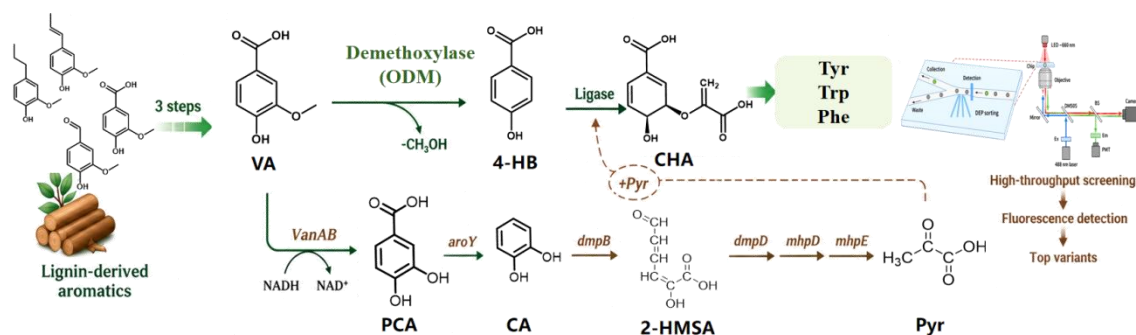


Figure 1. Overall strategy for high-value valorization of lignin-derived aromatics through non-natural pathway engineering and high-throughput evolution.

2. Result

2.1 Development and Functional Validation of a Whole-Cell Enzymatic Cascade for Vanillic Acid Production from Lignin

2.1.1 Construction of the lignin to Vanillic Acid Pathway in *E.coli*

To valorize products obtained from the electrochemical depolymerization of lignin, a whole-cell cascade was designed for the conversion of 4-propylguaiaicol (4-PG) to VA. The proposed pathway consists of three sequential reactions: Progo converts 4-PG to isoeugenol^[8,9], Nov1f converts isoeugenol to vanillin^[9], and vanillin dehydrogenase (Vdh) converts vanillin to VA^[16]. NADHoxidase (Nox) was included to regenerate NAD⁺ for the Vdh-catalyzed reaction^[16] (**Figure 2A**).

Against the backdrop of electrochemical depolymerization of lignin, the non-natural pathway was constructed. Enzyme activity was evaluated by whole-cell biocatalysis, with substrate consumption and product formation analyzed by HPLC (**Figure 2B**). SDS-PAGE was performed to assess recombinant protein expression at the expected molecular weight.

2.1.2 Validation of catalytic activities of Progo, Nov1f and Vdh

Before assembling the cascade, the catalytic activities of enzymes were verified individually. The strain expressing Progo produced isoeugenol from 4-PG with a yield of 6.08% after 3 h which is the rate-limiting enzyme of the overall reaction, whereas the Nov1f expressing strain converted isoeugenol to vanillin with a yield of 86.8% after 3 h as the highest yield. To examine Vdh activity independently, NAD⁺ was added to the reaction mixture which is equimolar to the substrate vanillin: the VA yield reached 85.5% after 6 h. These results confirmed activity at each step and suggested that conversion of 4-PG by Progo is the rate-limiting enzyme of the overall pathway.

2.1.3 Construction of the dual-enzyme module from 4-PG to vanillin and cofactor self-sufficient module of Vdh

To divided the 4 enzymes into 2 modules, pYB1a-*progo-nov1f* and pRB1k-*vdh-nox*, Nox oxidates NADH to supports NAD⁺ for Vdh, were constructed. The replication origin and resistance marker of the pYB1a-*vdh-nox* plasmid were changed to allow the two plasmids to be maintained and selected in the same host. The Progo-Nov1f module produced vanillin from 4-PG with a yield of 2.68% after 1 h (**Figure 2C**). For the Vdh-Nox module, which shows a VA yield of 35.3% in 6 h (**Figure 2D**).

2.1.4 Validation of Multi-enzyme Cascade System

To verify the complete catalytic activity of the whole muti-enzyme cascade system,

the 2 modules were co-transformed into BW25113 from 4-PG to VA. HPLC analysis showed a product peak consistent with the VA standard in the induced reaction, and the reported VA yield was 1.62% (**Figure 2E**). The formation of VA supports the feasibility of the whole-cell cascade. Although the overall yield remained lower than those of the individual downstream reactions, this result demonstrates that the engineered multi-enzyme system can achieve the complete conversion from 4-PG to VA and provides a solid foundation for further enhancement of cascade efficiency.

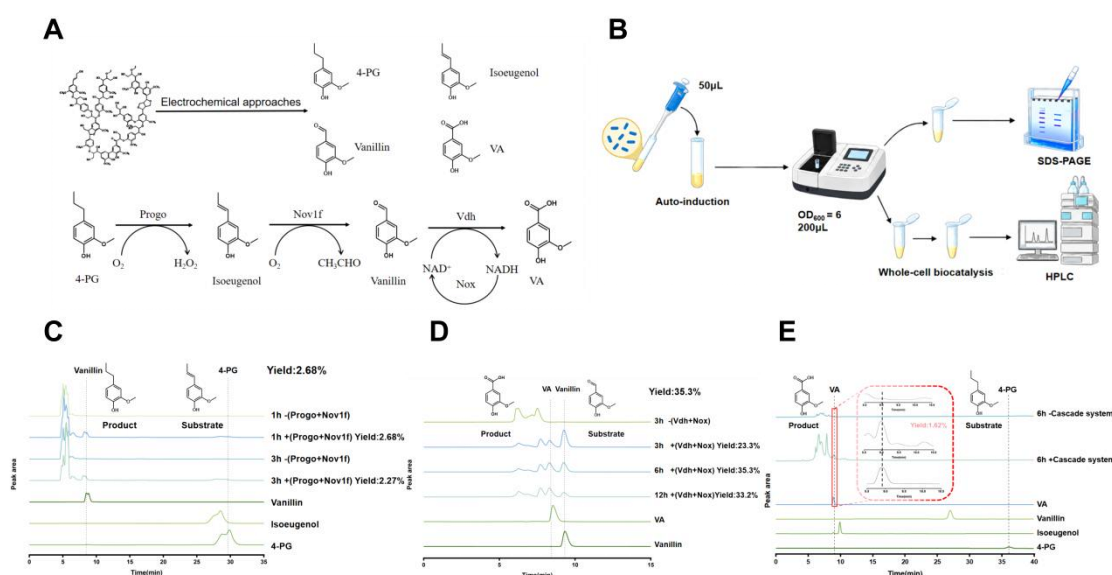


Figure 2. Design and evaluation of the whole-cell cascade for VA production from lignin-derived 4-PG.

(A) Proposed non-natural pathway involving Progo, Nov1f, Vdh, and Nox. (B) Workflow for cell induction, preparation, whole-cell biocatalysis, SDS-PAGE analysis, and HPLC detection to evaluate enzyme. (C) HPLC analysis of vanillin production from 4-PG by the Progo-Nov1f module. (D) HPLC analysis of VA production from vanillin by the Vdh-Nox module. (E) HPLC analysis of VA production from 4-PG by the complete cascade.

2.2 Biosensor-Assisted Screening and Directed Evolution of *odm* Variants

2.2.1 Construction of the Vanillic Acid-to-4-HB Pathway

To establish the key demethoxylation step, an ODM-mediated pathway was constructed for the conversion of VA to 4-HB^[17] (**Figure 3A**). Because the low catalytic activity of ODM limits this conversion, directed evolution was employed to improve its activity. To enable efficient screening of a large mutant library, a whole-cell biosensor was further introduced to couple 4-HB production to selectable cellular phenotypes.

2.2.2 P_{obR}-Based Biosensor for *odm* Screening

A 4-HB-responsive P_{obR}-based biosensor (BWΔ*codA*-gRB2k-*pobR*^{WT}-P_{pobA}*2-*mCherry-codA-Cmr*) was constructed, comprising the transcription factor P_{obR}, its cognate promoter P_{pobA}, and the downstream reporters *mCherry* and *Cmr*^[18] (**Figure 3B**). At low ODM activity, limited 4-HB production maintains P_{obR}-mediated transcriptional repression, resulting in weak fluorescence and poor Cmr. By contrast, highly active *odm* variants produce more 4-HB, thereby relieving P_{obR}-mediated repression and activating *mCherry* and *Cmr* expression, which leads to stronger fluorescence and enhanced Cmr. Thus, ODM catalytic activity can be translated into readily detectable and selectable cellular outputs.

2.2.3 Characterization of the Biosensor

The biosensor exhibited a clear 4-HB-dependent response: normalized mCherry fluorescence increased markedly with increasing 4-HB concentrations and gradually approached a plateau at higher concentrations (**Figure 3C**). Consistently, exogenous 4-HB enhanced bacterial growth across different Cmr concentrations (**Figure 3D**). These results demonstrate that the biosensor effectively couples 4-HB levels to both fluorescence output and growth-based selection.

2.2.4 Screening and Directed Evolution of ODM Variants

An *odm* mutant library was first constructed by error-prone PCR, followed by the introduction of the PobR-based biosensor for high-throughput screening (BWΔ*codA*-gRB2k-*pobR*^{WT}-P_{pobA}*2-*mCherry-codA-Cmr*/pYB1a-*odm*). Following biosensor-assisted screening, with parallel screening of the control (DH5α-pYB1a-*eGFP*; BWΔ*codA*-gRB2k-*pobR*^{WT}-P_{pobA}*2-*mCherry-codA-Cmr*; BWΔ*codA*), 29 candidate clones from mutant library A were selected for HPLC validation (**Figure 3E**). Among them, 20 variants exhibited higher catalytic activity than WT ODM, demonstrating effective enrichment of improved variants. Notably, variant 1-20-7 showed the highest catalytic activity, reaching 43.75-fold that of WT ODM in the primary screen and remaining at 29.28-fold that of the WT upon secondary screening. Further HPLC analysis confirmed the formation of the expected catalytic product (**Figure 3F**). Collectively, these results demonstrate that the PobR-based screening platform enables efficient identification of *odm* variants with

substantially enhanced catalytic activity.

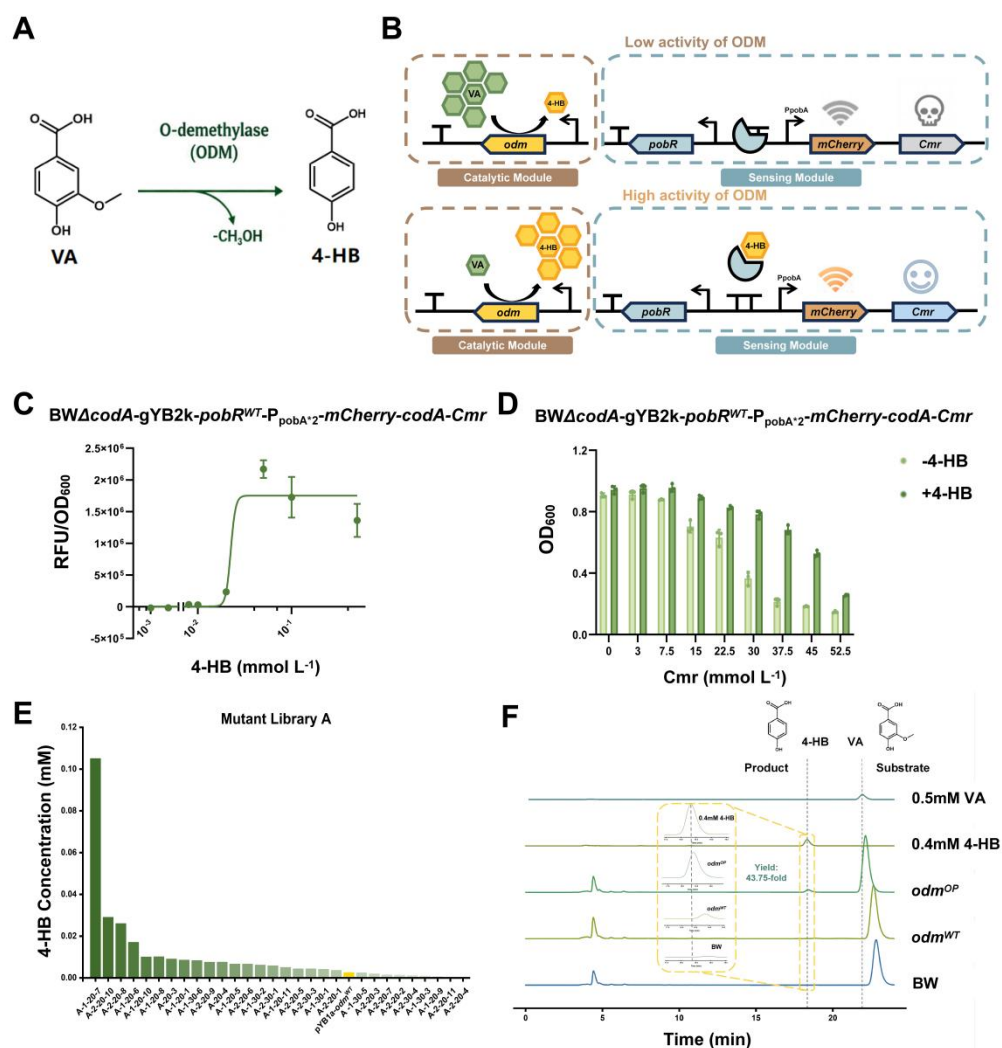


Figure 3. Development and application of a PobR-based biosensor for *odm* variant screening and HPLC activity validation.

(A) The 4-hydroxybenzoic acid biosynthesis pathway. VA; 4-HB. (B) Working principles of the constructed plasmids to screen for clones with high 4-HB production. Low ODM activity yields limited 4-HB from VA; *PobR* repressor remains bound to the *P_{pobA}* to silence *mCherry* and *Cmr*, leading to poor cell growth in *Cmr* medium. High ODM activity produces abundant 4-HB, which binds *PobR*, relieves transcriptional repression, and induces expression of *mCherry* (red fluorescence) and *Cmr* for cell survival. *pobR*: the gene encoding the allosteric transcription factor *PobR* protein that specifically responds to 4-hydroxybenzoic acid. *P_{pobA}*: a promoter that specifically binds the *PobR* complex to control downstream reporter gene expression. (C) Graph of the correlation between 4-HB concentration in the medium and the normalized fluorescence intensity of the biosensor strain, defined as red fluorescent protein (RFP) signal divided by optical density at 600 nm (OD₆₀₀). (D) Growth characterization of the biosensor strain under gradient *Cmr* concentrations, with or without exogenous 4-hydroxybenzoate (4-HB). (E) Catalytic activity of

variants from mutant library A after biosensor primary screening, quantified via HPLC. Among the 29 clones selected for detection in the first round of screening, 20 showed higher enzymatic activity than the WT (labeled in green color). (F) HPLC analysis of catalytic products from *odm* mutants.

2.3 Vanillic Acid is Transformed into Pyruvate via Enzyme

VA is converted to Pyr via a six-step enzymatic cascade. VA is first converted to PCA by the VanAB O-demethylase from *Pseudomonas putida*, followed by decarboxylation to CA by the AroY decarboxylase from *Klebsiella pneumoniae*^[19,20]. CA is then converted to 2-HMSA by the DmpB catechol 2,3-dioxygenase and subsequently to 2-HTDA by the DmpD hydrolase, both from *Pseudomonas* sp. CF600^[21,22]. Finally, 2-HTDA is converted to HOPA by the MhpD hydratase and HOPA to Pyr by the MhpE aldolase, both from *Escherichia coli* K-12^[23,24](**Figure 4A**).

The pathway was divided into module1 and module2. The module1 was plasmid pLB1s-*vanA-vanB-aroY-dmpD*, and the module2 was plasmid pYB1a-*dmpB-mhpD-mhpE*. Two plasmids jointly responsible for the transformation from CA to Pyr(**Figure 4B**). The Pyr-responsive growth-based sensor strain BWΔ*gldAΔmaeBΔpykF* was constructed for evaluating the function of the VA-to-Pyr pathway.

2.3.1 Validation of the Pyruvate Sensor Strain and Its Growth Response to Exogenous Pyruvate

To detect whether the whole pathway successfully produces Pyr, a Pyr-deficient sensor strain, BW Δ *gldA* Δ *maeB* Δ *pykF*, was constructed. After knocking out these three genes, the strain cannot synthesize sufficient Pyr on its own and must rely on exogenous Pyr for normal growth^[9]. Therefore, when the whole pathway successfully produces Pyr, the sensor strain can resume growth, coupling Pyr production to bacterial growth(**Figure 4B**).

To validate the response of the sensor strain to exogenous Pyr, four sodium Pyr concentration gradients (0, 1, 2, and 3 g/L) were set up, and growth curves were monitored in M9 medium. The results showed that the growth rate of the sensor strain increased significantly with increasing exogenous sodium Pyr concentration. The 3 g/L group grew fastest, followed by the 2 g/L group, then the 1 g/L group, and the 0 g/L group grew slowest. This indicated that the sensor strain can effectively respond to exogenous Pyr, and within a certain concentration range, the growth rate is positively correlated with Pyr concentration. In addition, in the absence of exogenous sodium Pyr, the growth rate of the sensor strain was significantly lower than that of WT BW, further confirming its Pyr-deficient nature(**Figure 4C**).

2.3.2 The Pyruvate Sensor Confirms Whole-Pathway Functionality Under Induced Conditions with Vanillic Acid

This section directly uses the sensor from Section 2.3.1. to validate whether the whole pathway can produce Pyr. The whole-pathway plasmids

BW-pLB1s-*vanA-vanB-aroY-dmpD*/pYB1a-*dmpB-mhpD-mhpE* were introduced into the sensor strain, and set up four experimental groups. The results showed that the group of ara induction with VA supplementation grew fastest and was significantly better than the other three groups; the group of ara non-induction with VA supplementation was second; the group of ara induction and ara non-induction without VA supplementation grew slowest. This indicated that under induced conditions with VA addition, the whole pathway can supply Pyr to the sensor and promote its growth(**Figure 4D**).

To further improve the conversion efficiency of the pathway, the RBS of the three genes *DmpB*, *MhpD*, and *MhpE* was mutated, and an RBS mutant library was constructed. The strategy was to perform random mutation of the RBS, construct a mutant library, and then combine biosensor-assisted screening with HPLC to obtain strains with higher conversion efficiency(**Figure 4E**).

The growth curves of the RBS mutant group and the WT group were compared. The results showed that at 0, 14, 19, 23, 38, and 43 h, the OD₆₀₀ values of the two groups were similar, with the RBS mutant group slightly higher at some time points(**Figure 4F**).

2.3.3 Establishment of a Pyruvate Sensor-Based Screening Workflow for RBS

Mutant Libraries

The RBS mutant library was successfully constructed, and a screening workflow based on the Pyr sensor was established^[16]. This workflow couples the pathway activity changes caused by RBS mutations to bacterial growth, enabling high-throughput screening of strains with higher conversion efficiency. Currently, this library is being further screened and validated to obtain strains with improved performance.

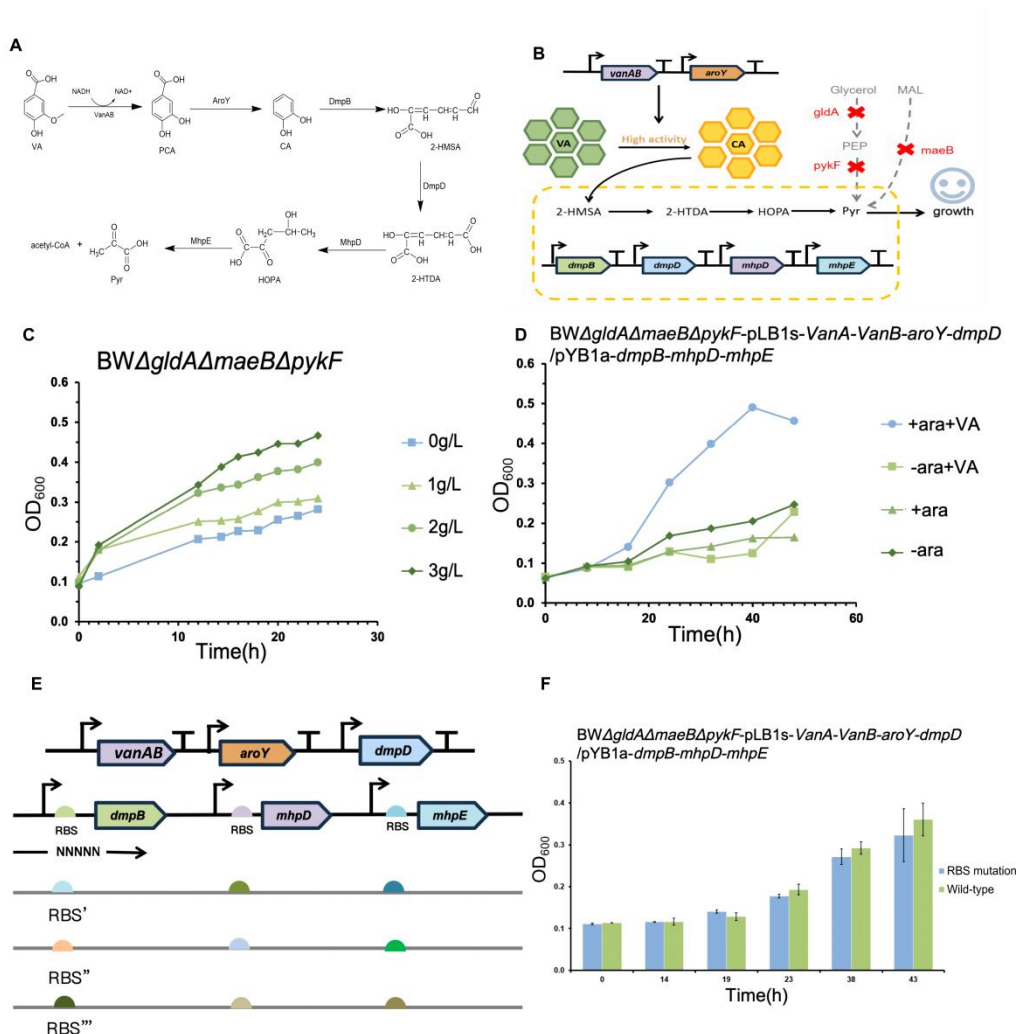


Figure 4. Development and Applications of Pyr Biosensor Validation and RBS Library Construction.

(A) The VA-to-Pyr pathway. (B) Principle of the Pyr-deficient sensor strain. (C) Growth response of the Pyr sensor to exogenous sodium Pyr. (D) Whole-pathway validation using the Pyr sensor. (E) RBS mutation and biosensor-assisted screening strategy. (F) Growth comparison between RBS mutant and WT strains.

2.4 Hardware-Assisted Microfluidic Screening

To address the screening bottleneck created by large mutant libraries^[25,26], a hardware-assisted microfluidic platform was constructed, integrating droplet generation, fluorescence detection, photon counting, and dielectrophoretic (DEP) sorting^[27,28].

2.4.1 Construction and Functional Validation of the Microfluidic Screening

Platform

Bacterial suspensions were diluted, encapsulated into water-in-oil droplets, incubated for 16 h, and introduced into the detection and sorting stage (**Figure 5A**). A camera provided droplet imaging, whereas the PMT converted fluorescence into photon counts. Signals were transferred through CH297 to the computer, and sorting commands were delivered through PYNQ-Z2, the function generator (AFG) and high-voltage amplifier (HVA) to the DEP electrodes (**Figure 5B**).

For eGFP detection, the optical path used 488 nm excitation with the indicated filters and DM505 (**Figure 5C**). FPGA counts increased linearly across seven tested input frequencies ($N = 16f$, $R^2 = 1.000$; **Figure 5D**). In 100 end-to-end electrical tests, 93 responses were successful; their latency was 25.359 ± 3.073 ms (mean $\pm$ SD; **Figure 5E**). eGFP samples showed repeated high-count peaks and net counts near 1,000, whereas uninduced controls remained near the lower baseline and about 150 net counts (**Figure 5F,5G**). The fluorescence-detection module can be adapted to the

mCherry reporter used in the PobR-based biosensor, providing a route toward integration with the *odm* screening workflow.

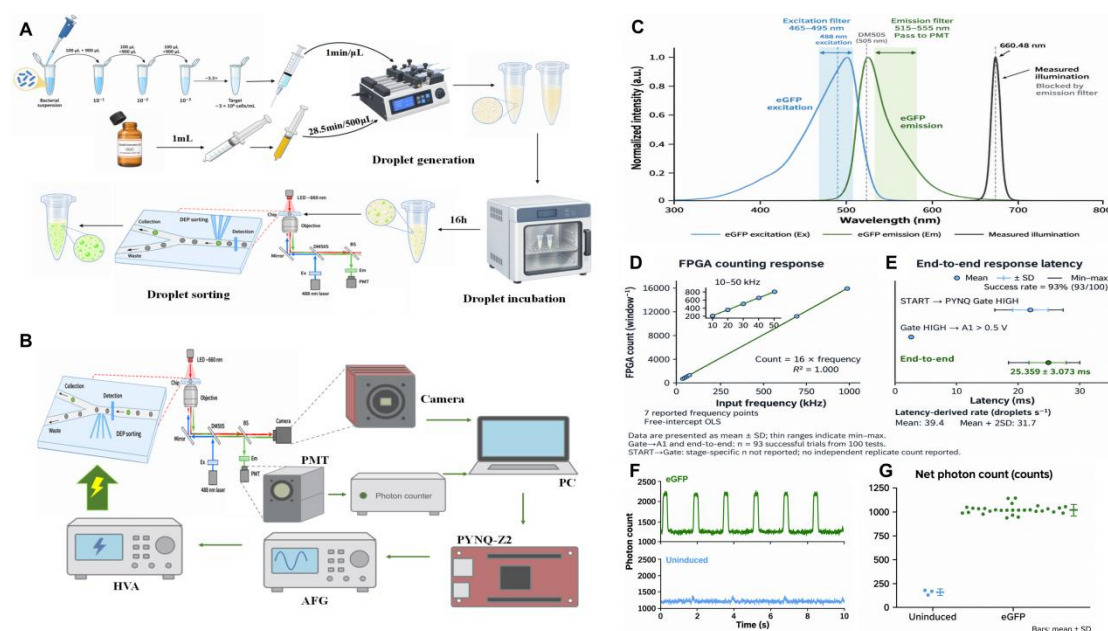


Figure 5. Workflow and functional validation of the microfluidic screening platform.

(A) Workflow from sample preparation to DEP sorting. (B) Hardware architecture. (C) Optical configuration. (D) FPGA counting response. (E) Electrical response latency. (F) Photon-count traces. (G) Net photon-count distributions.

2.4.2 Low-Cost Hardware Integration and Self-Developed Control Software

The physical platform was assembled and a dedicated control environment was developed (**Figure 6A, 6B**). The software contains camera observation, PMT photon counting, PYNQ gate control and syringe-pump fluid delivery modules (**Figure 6C-F**). Photon counts are read through CH297, evaluated using a threshold and hysteresis, and converted into HIGH/LOW commands for PYNQ. Actuation then proceeds through AFG, HVA and DEP.

Operational videos provided direct visual validation. Representative sorting frames at 1.507 s (HIGH) and 2.507 s (LOW) show periodic gate-state switching (**Figure 6G**), while frames at 0 and 1 s show continuous droplet formation (**Figure 6H**).

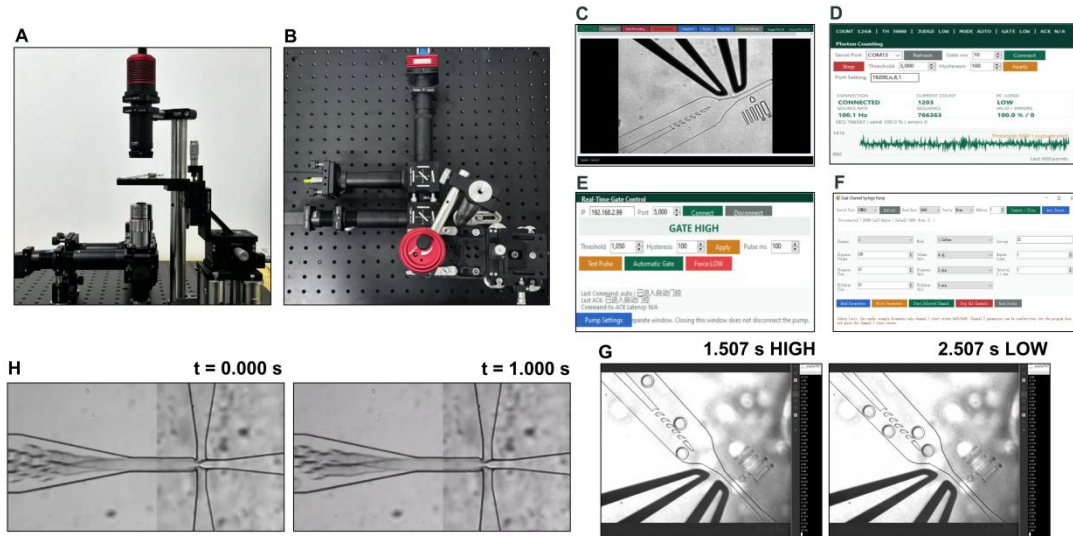


Figure 6. Engineering implementation and operational demonstration.

(A,B) Device photographs. (C) Camera interface. (D) PMT interface. (E) PYNQ interface. (F) Pump interface. (G) HIGH/LOW switching frames. (H) Droplet-generation frames.

The eight listed hardware categories had an estimated total cost of CNY 61,851 (approximately £6.9 k), with a recorded development period of five months. The estimate covers optics, imaging, detection, control and pumping hardware but excludes the PC, chips and consumables. Modules were designed to be replaceable and upgradable, and the control software and build guide are planned for public release through GitHub or another open repository.

Table 1. Estimated hardware cost of the microfluidic screening platform.

Component	Model	Cost (GBP)
HVA	ATA-2081	2,717.8
Optics/optomechanics	-	2,218.8
Syringe pump	LSP02-3B	741.8
AFG	DG1022Z	389.7
Camera	MUS40M-G	358.2
PMT	H10682	281.9
Photon counter	CH297	169.2
FPGA	PYNQ-Z2	97.7
Total	-	6,975.0

All additional experimental details, including primer sequences, plasmids, strains, and other relevant materials, are provided in the Supplementary Information.

3. Discussion

The major advantage of this study lies in establishing a completely non-natural VA-to-4-HB demethoxylation connection and rendering this highly challenging and previously unestablished biocatalytic transformation enzymatically accessible. Existing VA bioconversion remains constrained by native metabolism, with O-demethylation predominantly directing VA toward PCA catabolism^[11,12]. In contrast, direct VA-to-4-HB demethoxylation requires cleavage of a stable aromatic

C-O bond, and no characterized enzyme has been reported for this reaction^[29]. Here, this missing enzymatic connection was established de novo, and directed evolution increased ODM activity 29.28-fold over the WT. Coupling this breakthrough with the established 4-HB-to-CHA module redirects lignin-derived VA from native degradation toward CHA, establishing a completely non-natural VA-to-4-HB-to-CHA biosynthetic route and opening a previously inaccessible direction for high-value lignin valorization.

Furthermore, the multi-enzyme cascade successfully funneled lignin-derived aromatics into VA, with Progo identified as the rate-limiting enzyme and Nov1f and Vdh achieved high single-step conversion rates. Previous studies have mostly validated individual enzymes. One report concerning the combined validation of Progo and Nov1f adopted a one-pot strategy suffering from certain limitations^[10]. Dual-enzyme modules and cascade were validated, demonstrating efficient convergence of aromatic carbons into VA for enhanced utilization and economic value. In addition, A non-natural VA-to-Pyr pathway was constructed and validated, providing endogenous Pyr for the 4-HD-to-CHA step, thereby greening the overall lignin valorization route and enhancing its sustainability.

However, this study did not stop there. Beyond biological screening, this study developed a microfluidic platform to support faster, more accurate screening. Despite complex instrumentation requirements^[15], an undergraduate team integrated

fluorescence detection and DEP sorting within five months for approximately CNY 62,000 in hardware. Equipment reuse and AI-assisted development of four device control programs and one integrated control platform combined affordability with engineering autonomy, enabling independent maintenance and upgrades.

Taken together, these advances establish an integrated framework spanning aromatic-carbon funneling, completely non-natural pathway construction, sustainable metabolic engineering, and autonomous high-throughput screening. By transcending native metabolic constraints from both biological and engineering perspectives, this study opens a new route for redirecting lignin-derived aromatic carbon toward high-value biomanufacturing.

Despite demonstrating overall feasibility, current module-level yields remain insufficient for practical application, and the individual modules have not yet been integrated into a complete lignin-to-CHA pathway. Given the limited research period and parallel development of multiple modules, this study prioritized independent proof-of-concept validation. Future efforts will advance from module-level feasibility to pathway-level optimization, integrating flux balancing, enzyme engineering, and iterative evolution toward a robust lignin-to-CHA biomanufacturing platform.

4. Materials and Methods

4.1 Chemicals, Strains and Materials

4.1.1 Compounds and Standards

A variety of key intermediates including 4-PG, Isoeugenol, vanillin, VA, 4-HB, CA and Pyr were involved. All these compounds were purchased from Aladdin Biochemical Technology Co., Ltd. Other reagents used in the experiments were conventional molecular biology reagents. In this study, a series of standard solutions of 4-PG, Isoeugenol, vanillin, VA, 4-HB with concentration gradients (0-1 mM) were prepared. High-performance liquid chromatography (HPLC) was employed to obtain the standard curves for them.

4.1.2 Media and Culture Conditions

During the cultivation of *E. coli* strains, LB, M9 and ZYM5052 media were used.

4.1.3 Strains and Plasmids

The main strain employed in this work was DH5 α , BW25113, BW Δ codA; the main plasmid includes BW Δ codA-gRB2k-pobR^{WT}-P_{pobA*2}-mCherry-codA-Cmr/pYB1a-odm, DH5 α -pYB1a-odm^{WT}, DH5 α -pYB1a-ccdB, BW Δ codA-gRB2k-pobR^{WT}-P_{pobA*2}-mCherry-codA-Cmr/pYB1a-odm, BW-pLB1s-vanA-vanB-aroY-dmpD, BW-pYB1a-dmpB-mhpD-mhpE, BW-pLB1s-vanA-vanB-aroY-dmpD/pYB1a-dmpB-mhpD-mhpE, and the Pyr-deficient sensor strain BW Δ gldA Δ maeB Δ pykF.

The remaining materials included recombinant gene plasmid vectors and elements related to whole-cell biosensors. Primers used for plasmid construction were synthesized by Ruibo Biology (Harbin, China)^[30].

4.1.4 Gene Synthesis and Plasmid Construction

Genes including *Progo*, *NovIf*, *Vdh*, *Nox*, *VanAB*, *aroY*, *dmpB*, *dmpD*, *mhpD*, and *mhpE* were used in this study. Plasmid construction was performed using the Gibson Assembly method (Gibson Assembly, 2 × CE Mix, C117)^[31], and cloning was completed using the Gibson Assembly Cloning Kit (New England Biolabs, NEB).

4.2 Construction of Recombinant Strains and the Multi-Enzyme Cascade System

Using *E. coli* BW25113 as the host, recombinant strains expressing *Progo*, *NovIf*, *Vdh*, and BW-pYB1a-*vdh-nox* were constructed, along with a co-expression strain, BW-pYB1a-*progo-novIf*. To allow the two plasmids to coexist in the same host, the replication origin and antibiotic selection marker of the original BW-pYB1a-*vdh-nox* plasmid were modified to obtain pRB1k-*vdh-nox*. The two plasmids carry out the first two steps and the terminal oxidation step of the cascade, respectively. Plasmid construction was verified by colony PCR and restriction enzyme digestion.

Using *E. coli* BW25113 as the host, recombinant strains expressing *vanA*, *vanB*, *aroY*, *dmpB*, *dmpD*, *mhpD*, and *mhpE* were constructed. A co-expression strain,

BW-pLB1s-*vanA-vanB-aroY-dmpD*/pYB1a-*dmpB-mhpD-mhpE*, was constructed for the whole pathway. To allow the two plasmids to coexist in the same host, compatible replication origins and different antibiotic selection markers were used. The two plasmids carry out the conversion of VA to CA and CA to Pyr, respectively. Additionally, a Pyr-deficient sensor strain, BW Δ *gldA* Δ *maeB* Δ *pykF*, was constructed for pathway validation^[32].

4.3 Construction and Characterization of the Biosensor

4.3.1 Construction of the BWcodA-gRB2k-pobR^{WT}-P_{pobA*2}-mCherry-CD-Cmr-Based

This study accomplished the design and construction of the BW Δ codA-gRB2k-pobRWT-PpobA*2-mCherry-codA-Cmr. The sensing element originated from the IDEC project conducted in the previous three years, and this study adopted and inherited this element system[33].

4.3.2 Characterization of the 4-HB-Responsive Selection System

Activate the strain. The prepared 1×M9 medium was used to dilute the 100 mmol/L 4-HB stock solution to a final concentration of 1 mmol/L. 4-HB-free system in 96-well plate: 1 μ L bacterial culture; 0.2 μ L Ampicillin.

Table 2. Characterization reaction system of**BWΔ*codA*-gRB2k-*pobR*^{WT}-P_{pobA*2}-*mCherry-codA-Cmr* in response to 4-HB.**

4-HB Concentration (mmol/L)	1 mmol/L 4-HB (μL)	1×M9 (μL)
0	0	200
0.001	20	180
0.002	40	160
0.008	160	40
0.01	2	198
0.02	4	196
0.05	10	190
0.1	20	180
0.5	100	100
0.6	120	80

The fluorescence intensity and OD₆₀₀ values of the samples were then measured using a microplate reader in 12 h, and graphs were plotted based on the acquired data.

4.3.3 Determination of Chloramphenicol Selection Conditions

Activate the strain. Reagent preparation: 0.6 mmol/L was selected as the optimal stable concentration of 4-HB. The 100 mmol/L 4-HB stock solution was diluted to 10 mmol/L. A 0.3 mg/mL Cmr stock solution was prepared, and Cmr concentration gradients were set as follows.

Table 3. Reaction System for Characterizing the Cmr Response of *BWΔcodA-gRB2k-pobR^{WT}-P_{pobA}*2-mCherry-codA-Cmr*.

Cmr		Addition of 4-HB (0.6 mmol/L)			No addition of 4-HB	
Concentration						
Gradient		Cmr (μL)	4-HB (μL)	LB (μL)	Cmr (μL)	LB (μL)
(mg/L)						
1	0	0	12	188	0	200
2	3	2	12	186	2	198
3	7.5	5	12	183	5	195
4	15	10	12	178	10	190
5	22.5	15	12	173	15	185
6	30	20	12	168	20	180
7	37.5	25	12	163	25	175
8	45	30	12	158	30	170
9	52.5	35	12	153	35	165

After incubation, the OD values were measured using a microplate reader, and plots were generated based on the detected data.

4.3.4 Pyruvate Sensor Characterization

Exogenous sodium Pyr gradient experiments were set at 0, 1, 2, and 3 g/L. The 96-well plate system used 200 μ L per well, cultured at 30 °C, with OD₆₀₀ measured every 30 min using a microplate reader. Program settings: temperature 30 °C, orbital shaking at 216 rpm, wavelength 600 nm.

4.4 Construction of the Demethoxylase Mutant Library

4.4.1 Random Mutagenesis by Error-Prone PCR

Random mutagenesis was adopted to generate mutants. Briefly, random mutation was performed on the *odm* gene. An error-prone PCR technique was used to construct the mutant library. The PCR program was set as follows: initial denaturation at 98 °C for 5 min; followed by 30 cycles, each consisting of denaturation at 98 °C for 30 s, annealing at 57 °C for 30 s, and extension at 72 °C for 30 s; final extension at 72 °C for 5 min, and then holding at 25 °C.

4.4.2 Construction of the Mutant Plasmid Library

Vector PCR amplification was performed using DH5 α -pYB1a-*ccdB* as the template, and the linearized vector fragment was recovered. The purified vector fragment and target fragment were assembled via Gibson homologous recombination (C117

enzyme). After preparation of the reaction system, sufficient incubation was carried out to complete assembly, yielding abundant recombinant ligation products. The assembly conditions were set as follows: 50 °C for 30 min, followed by holding at 25 °C.

The ligation products were transformed into 22 aliquots (100 µL each) of *E. coli* DH5α competent cells to obtain recombinant strains. Following the reported protocol^[32], the transformed cell suspensions were spread onto LB solid agar plates containing 100 mg/L ampicillin, and incubated at 37 °C for 12 h. According to scraping the colony and shake-culture, plasmids were extracted to generate the mutant plasmid library of *odm*. The capacity of this plasmid library was approximately 1.61×10^5 clones.

4.4.3 Transformation and Mutant Library Establishment

The plasmid library was transformed into the host strain

BWΔ*codA*-gRB2k-*pobR*^{WT}-P_{pobA}*2-*mCherry-codA-Cmr*, which harbors the 4-HB-responsive screening system. Two rounds of library construction were performed. The first library capacity was approximately 50,000 clones and the second library capacity was approximately 66,000 clones.

4.4.4 Evaluation of Mutant Library Quality

The capacity of the constructed plasmid library was quantified, and the mutation frequency as well as the number of mutation sites per monoclonal of the library were determined. In this study, there were 1-3 mutation sites within the approximately 1156 base pair (bp) target fragment for each monoclonal, and the mutation frequency was approximately 0.2%.

4.5 RBS Library Construction and Screening

Random mutagenesis was performed on the RBS sequences of the three genes *DmpB*, *MhpD*, and *MhpE*. Degenerate primers (NNNNN) were designed, and RBS mutations were introduced by PCR amplification to construct an RBS mutant library. The mutant library was combined with the Pyr sensor, and strains with improved conversion efficiency were screened by growth advantage^[34].

The RBS mutant library was transformed into *BWΔgldAΔmaeBΔpykF/pLB1s-vanAB-aroY-dmpD* competent cells and plated on LB plates containing appropriate antibiotics. Single colonies were inoculated into M9 medium, cultured at 30°C, and OD₆₀₀ was monitored periodically. Growth curves of the RBS mutant group and WT group were compared.

4.6 Chloramphenicol-Resistance-Based Screening of Demethoxylase Mutants

4.6.1 Preliminary Selection of the Mutant Library

The mutant library was subcultured into 5 mL of 1×M9 medium at an inoculation ratio of 1%^[33]. The screening system was prepared as follows:

Table 4. *odm* Screening System

Components	20 mg/L Cmr		30 mg/L Cmr	
	WT(pobR) Group	Mixed Group	WT(pobR) Group	Mixed Group
Antibiotics (μL)	5	5	5	5
5×M9 (mL)	1	1	1	1
20% Glycerol (μL)	500	500	500	500
MgSO ₄ (μL)	10	10	10	10
CaCl ₂ (μL)	0.5	0.5	0.5	0.5
20% L-Arabinose (μL)	50	50	50	50
Cmr (μL)	6.7	6.7	10	10
10 mM VA (μL)	1250	1250	1250	1250
double-distilled water (μL)	2122.8	2122.8	2119.5	2119.5
Bacterial Suspension (μL)	50	50	50	50

Cultivation was performed at 30 °C with shaking at 200 rpm for 12 h. When the bacterial suspension turned turbid (indicating substantial cell proliferation), subculture

was carried out again, and this transfer procedure was repeated for three rounds in total.

4.6.2 Multi-Round Enrichment of High-Activity Variants

After multiple rounds of screening, candidate mutants with high activity were gradually enriched in the system^[34].

4.6.3 Isolation of Candidate Demethoxylase Variants

After screening, streak-plate isolation and cultivation were performed. Single colonies were picked to obtain candidate mutant strains.

4.7 HPLC Analysis and Validation

4.7.1 Preparation of Standards and Calibration Curves

VA and 4-HB standard solutions were prepared at concentration gradients of 0 mM, 0.2 mM, 0.4 mM, 0.6 mM, 0.8 mM and 1 mM.

4.7.2 Sample Preparation for HPLC Analysis

After strain activation, induction and whole-cell catalysis, the sample is prepared for HPLC analysis^[35].

4.7.3 HPLC Conditions

An Agilent 1100 series high-performance liquid chromatography system was used for analysis equipped with an Agilent XB-C18 column (4.6×250 mm, 5μm). The detailed detection parameters were set as follows:

a. 4-PG, Isoeugenol, vanillin and VA: detection wavelength, 280 nm; mobile phase, a mixture of methanol and 0.1% (w/w) glacial acetic acid solution at a volume ratio of 65:35; injection volume, 10 μL; flow rate, 0.4 mL/min^[34].

b. 4-HB and VA: detection wavelength, 254 nm; mobile phase, a mixture of methanol and 0.1% (w/w) aqueous phosphoric acid solution at a volume ratio of 25:75; injection volume, 5 μL; flow rate, 0.5 mL/min^[35].

4.7.4 Validation of Improved odm Variants

Parallel bioconversion assays of candidate mutants and WT enzyme were performed under identical reaction conditions. The consumption of substrate VA and production of 4-HB were quantified by HPLC. The bioconversion efficiencies were statistically compared to finally screen and verify mutants with genuinely improved capacity for VA to 4-HB conversion.

4.8 Microfluidic Droplet Generation and Dielectrophoretic Sorting

4.8.1 Droplet Generation and Incubation

The aqueous phase, oil phase, and collection tubing were connected to a PDMS droplet-generation chip^[15]. The aqueous and oil phases were driven separately using syringe pumps. The flow rate of the aqueous phase was set to approximately 1.5 $\mu\text{L}/\text{min}$, while the oil phase was injected at a rate of 500 μL per 28.5 min. After stable droplet generation was achieved, the droplets were collected and incubated at 30 °C for 16 h.

4.8.2 Hardware Connection and Testing

The output of the photomultiplier tube (PMT) was connected to the CH297 photon counter, which was further connected to the experimental computer to establish the PMT-CH297-computer photon-counting acquisition pathway. The PYNQ-Z2 board was then connected to the experimental computer via an Ethernet cable, and communication between the board-side control program and the integrated control program was established.

Photon-counting tests were subsequently performed under dark-box conditions. The gate time of the CH297 was set to 10 ms, and the low background photon-count level in the dark box was measured and transmitted to the experimental computer to verify the response of the PMT-CH297 detection pathway under low-light conditions.

Before connecting the high-voltage system, an oscilloscope was connected to PMOD B Pin 1 of the PYNQ-Z2. The initial output state was confirmed to be GATE LOW,

and test pulses were generated to verify an approximately 3.3 V digital output. After testing, the output was returned to LOW.

The PYNQ-Z2 output was then connected to the external trigger input of a Rigol DG1022Z function generator. The waveform output from the function generator was first examined using an oscilloscope to confirm that its waveform, frequency, and amplitude were normal. The function generator was subsequently connected to an ATA-2081 high-voltage amplifier, and the amplified signal was applied to the DEP electrodes of the sorting chip. DEP sorting was performed using an approximately 8 kHz sinusoidal signal, with an actual high-voltage amplifier output of approximately 300-500 V_{pp}^[36].

4.8.3 Droplet Detection and Sorting

After incubation, the droplets were introduced into the microfluidic sorting chip for fluorescence detection, photon counting, and camera monitoring. Photon counts from the droplets were continuously recorded for approximately 5-10 min, and the sorting threshold was determined based on the photon-count results obtained during this period. Automatic gating was then enabled in the integrated control program^[27].

The photon count obtained during each 10 ms gate period was continuously compared with the sorting threshold. When the fluorescence signal did not reach the triggering threshold, droplets flowed through the default channel into the Waste outlet.

When the photon count reached the sorting threshold, DEP sorting was triggered, causing the target droplets to change their trajectories and enter the collection channel^[36,37]. After each triggering event, the system returned to its default state for subsequent droplet detection.

During sorting, camera images, CH297 photon counts, threshold-determination results, and sorting states were synchronously recorded to enable subsequent correlation of droplet motion with fluorescence detection and sorting events.

Declaration of competing interest

The authors declare no competing interests.

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